Chapter 3 Re engineering

Compiled reviewer · 1 lesson(s)

Lesson 1: Chapter 3 Re engineering

Chapter 4– Requirements Engineering

Lecture 1

Keywords: Chapter, Requirements Engineering Lecture

Topics covered

Functional and non-functional requirements

The software requirements document Requirements specification Requirements engineering processes Requirements elicitation and analysis Requirements validation Requirements management

- The software requirements document
- Requirements specification
- Requirements engineering processes
- Requirements elicitation and analysis
- Requirements validation
- Requirements management

Keywords: Topics, Functional, Requirements

Requirements engineering

The process of establishing the services that the customer requires from a system and the constraints under which it operates and is developed. The requirements themselves are the descriptions of the system services and constraints that are generated during the requirements engineering process.

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- The requirements themselves are the descriptions of the system services and constraints that are generated during the requirements engineering process.

Keywords: Requirements

What is a requirement?

It may range from a high-level abstract statement of a service or of a system constraint to a detailed mathematical functional specification. This is inevitable as requirements may serve a dual function May be the basis for a bid for a contract - therefore must be open to interpretation; May be the basis for the contract itself - therefore must be defined in detail; Both these statements may be called requirements.

- It may range from a high-level abstract statement of a service or of a system constraint to a detailed mathematical functional specification.
- This is inevitable as requirements may serve a dual function
- May be the basis for a bid for a contract - therefore must be open to interpretation;
- May be the basis for the contract itself - therefore must be defined in detail;
- Both these statements may be called requirements.

Keywords: What, Both

Types of requirement

User requirements

Statements in natural language plus diagrams of the services the system provides and its operational constraints. System requirements A structured document setting out detailed descriptions of the system's functions, services and operational constraints.

- Statements in natural language plus diagrams of the services the system provides and its operational constraints. Written for customers.
- System requirements
- A structured document setting out detailed descriptions of the system's functions, services and operational constraints. Defines what should be implemented so may be part of a contract between client and contractor.

Keywords: Types, User, Statements, Written, System, Defines

Functional and non-functional requirements

Functional requirements

Statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations. Domain requirements Constraints on the system from the domain of operation

- Statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations.
- May state what the system should not do.
- Non-functional requirements
- Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.
- Often apply to the system as a whole rather than individual features or services.
- Domain requirements
- Constraints on the system from the domain of operation

Keywords: Functional, Statements, Constraints, Often, Domain

Functional requirements

Describe functionality or system services.

Depend on the type of software, expected users and the type of system where the software is used. Functional system requirements should describe the system services in detail.

- Depend on the type of software, expected users and the type of system where the software is used.
- Functional user requirements may be high-level statements of what the system should do.
- Functional system requirements should describe the system services in detail.

Keywords: Functional, Describe, Depend

Functional requirements for the MHC-PMS

A user shall be able to search the appointments lists for all clinics.

The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day. Each staff member using the system shall be uniquely identified by his or her 8-digit employee number.

- The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
- Each staff member using the system shall be uniquely identified by his or her 8-digit employee number.

Keywords: Functional, Each

Requirements completeness and consistency

In principle, requirements should be both complete and consistent.

Complete They should include descriptions of all facilities required. In practice, it is impossible to produce a complete and consistent requirements document.

- Complete
- They should include descriptions of all facilities required.
- Consistent
- There should be no conflicts or contradictions in the descriptions of the system facilities.
- In practice, it is impossible to produce a complete and consistent requirements document.

Keywords: Requirements, Complete They, Consistent There

Non-functional requirements

These define system properties and constraints e.g. Constraints are I/O device capability, system representations, etc.

- These define system properties and constraints e.g. reliability, response time and storage requirements. Constraints are I/O device capability, system representations, etc.
- Process requirements may also be specified mandating a particular IDE, programming language or development method.
- Non-functional requirements may be more critical than functional requirements. If these are not met, the system may be useless.

Keywords: Constraints, Process

Non-functional requirements implementation

Non-functional requirements may affect the overall architecture of a system rather than the individual components. It may also generate requirements that restrict existing requirements.

- Non-functional requirements may affect the overall architecture of a system rather than the individual components.
- For example, to ensure that performance requirements are met, you may have to organize the system to minimize communications between components.
- A single non-functional requirement, such as a security requirement, may generate a number of related functional requirements that define system services that are required.
- It may also generate requirements that restrict existing requirements.

Non-functional classifications

Product requirements

process standards used, implementation requirements, etc. interoperability requirements, legislative requirements, etc.

- Requirements which specify that the delivered product must behave in a particular way e.g. execution speed, reliability, etc.
- Organisational requirements
- Requirements which are a consequence of organisational policies and procedures e.g. process standards used, implementation requirements, etc.
- External requirements
- Requirements which arise from factors which are external to the system and its development process e.g. interoperability requirements, legislative requirements, etc.

Keywords: Product, Requirements, Organisational, External

Goals and requirements

Non-functional requirements may be very difficult to state precisely and imprecise requirements may be difficult to verify. Verifiable non-functional requirement A statement using some measure that can be objectively tested.

- Non-functional requirements may be very difficult to state precisely and imprecise requirements may be difficult to verify.
- Goal
- A general intention of the user such as ease of use.
- Verifiable non-functional requirement
- A statement using some measure that can be objectively tested.
- Goals are helpful to developers as they convey the intentions of the system users.

Keywords: Goals, Goal, Verifiable

Usability requirements

Medical staff shall be able to use all the system functions after four hours of training. (Testable non-functional requirement)

- The system should be easy to use by medical staff and should be organized in such a way that user errors are minimized. (Goal)

Keywords: Usability, Medical, After, Testable, Goal

Key points

Requirements for a software system set out what the system should do and define constraints on its operation and implementation. Non-functional requirements often constrain the system being developed and the development process being used.

- Requirements for a software system set out what the system should do and define constraints on its operation and implementation.
- Functional requirements are statements of the services that the system must provide or are descriptions of how some computations must be carried out.
- Non-functional requirements often constrain the system being developed and the development process being used.
- They often relate to the emergent properties of the system and therefore apply to the system as a whole.

Keywords: Requirements, Functional, They

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Lecture 2

Keywords: Chapter, Requirements Engineering Lecture

The software requirements document

The software requirements document is the official statement of what is required of the system developers. Should include both a definition of user requirements and a specification of the system requirements.

- The software requirements document is the official statement of what is required of the system developers.
- Should include both a definition of user requirements and a specification of the system requirements.
- It is NOT a design document. As far as possible, it should set of WHAT the system should do rather than HOW it should do it.

Keywords: Should, WHAT

Agile methods and requirements

Many agile methods argue that producing a requirements document is a waste of time as requirements change so quickly. Methods such as XP use incremental requirements engineering and express requirements as ‘user stories’ (discussed in Chapter 3).

- Many agile methods argue that producing a requirements document is a waste of time as requirements change so quickly.
- The document is therefore always out of date.
- Methods such as XP use incremental requirements engineering and express requirements as ‘user stories’ (discussed in Chapter 3).
- This is practical for business systems but problematic for systems that require a lot of pre-delivery analysis (e.g. critical systems) or systems developed by several teams.

Keywords: Agile, Many, Methods, Chapter

Requirements specification

The process of writing don the user and system requirements in a requirements document.

User requirements have to be understandable by end-users and customers who do not have a technical background. System requirements are more detailed requirements and may include more technical information.

- User requirements have to be understandable by end-users and customers who do not have a technical background.
- System requirements are more detailed requirements and may include more technical information.
- The requirements may be part of a contract for the system development
- It is therefore important that these are as complete as possible.

Keywords: Requirements, User, System

Requirements and design

In principle, requirements should state what the system should do and the design should describe how it does this. In practice, requirements and design are inseparable A system architecture may be designed to structure the requirements; The system may inter-operate with other systems that generate design requirements; The use of a specific architecture to satisfy non-functional requirements may be a domain requirement.

- In principle, requirements should state what the system should do and the design should describe how it does this.
- In practice, requirements and design are inseparable
- A system architecture may be designed to structure the requirements;
- The system may inter-operate with other systems that generate design requirements;
- The use of a specific architecture to satisfy non-functional requirements may be a domain requirement.
- This may be the consequence of a regulatory requirement.

Keywords: Requirements

Natural language specification

Requirements are written as natural language sentences supplemented by diagrams and tables. Used for writing requirements because it is expressive, intuitive and universal.

- Requirements are written as natural language sentences supplemented by diagrams and tables.
- Used for writing requirements because it is expressive, intuitive and universal. This means that the requirements can be understood by users and customers.

Keywords: Natural, Requirements, Used

Guidelines for writing requirements

Invent a standard format and use it for all requirements.

Use language in a consistent way. Avoid the use of computer jargon.

- Use language in a consistent way. Use shall for mandatory requirements, should for desirable requirements.
- Use text highlighting to identify key parts of the requirement.
- Avoid the use of computer jargon.
- Include an explanation (rationale) of why a requirement is necessary.

Keywords: Guidelines, Invent, Avoid, Include

Problems with natural language

Lack of clarity

Requirements confusion Functional and non-functional requirements tend to be mixed-up. Requirements amalgamation Several different requirements may be expressed together.

- Precision is difficult without making the document difficult to read.
- Requirements confusion
- Functional and non-functional requirements tend to be mixed-up.
- Requirements amalgamation
- Several different requirements may be expressed together.

Keywords: Problems, Lack, Precision, Requirements, Functional, Several

Structured specifications

An approach to writing requirements where the freedom of the requirements writer is limited and requirements are written in a standard way. requirements for embedded control system but is sometimes too rigid for writing business system requirements.

- An approach to writing requirements where the freedom of the requirements writer is limited and requirements are written in a standard way.
- This works well for some types of requirements e.g. requirements for embedded control system but is sometimes too rigid for writing business system requirements.

Keywords: Structured

Form-based specifications

Definition of the function or entity.

Description of inputs and where they come from. Information about the information needed for the computation and other entities used.

- Description of inputs and where they come from.
- Description of outputs and where they go to.
- Information about the information needed for the computation and other entities used.
- Description of the action to be taken.
- Pre and post conditions (if appropriate).
- The side effects (if any) of the function.

Keywords: Form, Definition, Description, Information

Requirements engineering processes

The processes used for RE vary widely depending on the application domain, the people involved and the organisation developing the requirements. However, there are a number of generic activities common to all processes Requirements elicitation; Requirements analysis; Requirements validation; Requirements management.

- The processes used for RE vary widely depending on the application domain, the people involved and the organisation developing the requirements.
- However, there are a number of generic activities common to all processes
- Requirements elicitation;
- Requirements analysis;
- Requirements validation;
- Requirements management.
- In practice, RE is an iterative activity in which these processes are interleaved.

Keywords: Requirements, However

Requirements elicitation and analysis

Sometimes called requirements elicitation or requirements discovery.

Involves technical staff working with customers to find out about the application domain, the services that the system should provide and the system's operational constraints. May involve end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc.

- Involves technical staff working with customers to find out about the application domain, the services that the system should provide and the system's operational constraints.
- May involve end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc. These are called stakeholders.

Keywords: Requirements, Sometimes, Involves

Problems of requirements analysis

Stakeholders don't know what they really want.

Stakeholders express requirements in their own terms. Different stakeholders may have conflicting requirements.

- Stakeholders express requirements in their own terms.
- Different stakeholders may have conflicting requirements.
- Organisational and political factors may influence the system requirements.
- The requirements change during the analysis process. New stakeholders may emerge and the business environment may change.

Keywords: Problems, Stakeholders, Different, Organisational

Requirements elicitation and analysis

Software engineers work with a range of system stakeholders to find out about the application domain, the services that the system should provide, the required system performance, hardware constraints, other systems, etc.

- Stages include:
- Requirements discovery,
- Requirements classification and organization,
- Requirements prioritization and negotiation,
- Requirements specification.

Keywords: Requirements, Software, Stages

Process activities

Requirements discovery

Requirements classification and organisation Groups related requirements and organises them into coherent clusters. Prioritisation and negotiation Prioritising requirements and resolving requirements conflicts.

- Interacting with stakeholders to discover their requirements. Domain requirements are also discovered at this stage.
- Requirements classification and organisation
- Groups related requirements and organises them into coherent clusters.
- Prioritisation and negotiation
- Prioritising requirements and resolving requirements conflicts.
- Requirements specification
- Requirements are documented and input into the next round of the spiral.

Keywords: Process, Requirements, Interacting, Domain, Groups, Prioritisation, Prioritising

Problems of requirements elicitation

Stakeholders don't know what they really want.

Stakeholders express requirements in their own terms. Different stakeholders may have conflicting requirements.

- Stakeholders express requirements in their own terms.
- Different stakeholders may have conflicting requirements.
- Organisational and political factors may influence the system requirements.
- The requirements change during the analysis process. New stakeholders may emerge and the business environment change.

Keywords: Problems, Stakeholders, Different, Organisational

Key points

Requirements elicitation and analysis is an iterative process that can be represented as a spiral of activities – requirements discovery, requirements classification and organization, requirements negotiation and requirements documentation.

- The software requirements document is an agreed statement of the system requirements. It should be organized so that both system customers and software developers can use it.
- The requirements engineering process is an iterative process including requirements elicitation, specification and validation.

Keywords: Requirements

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Lecture 3

Keywords: Chapter, Requirements Engineering Lecture

Requirements discovery

The process of gathering information about the required and existing systems and distilling the user and system requirements from this information. Systems normally have a range of stakeholders.

- The process of gathering information about the required and existing systems and distilling the user and system requirements from this information.
- Interaction is with system stakeholders from managers to external regulators.
- Systems normally have a range of stakeholders.

Keywords: Requirements, Interaction, Systems

Stakeholders in the MHC-PMS

Patients whose information is recorded in the system.

Doctors who are responsible for assessing and treating patients. Medical receptionists who manage patients' appointments.

- Doctors who are responsible for assessing and treating patients.
- Nurses who coordinate the consultations with doctors and administer some treatments.
- Medical receptionists who manage patients' appointments.
- IT staff who are responsible for installing and maintaining the system.

Keywords: Stakeholders, PMS Patients, Doctors, Nurses, Medical

Stakeholders in the MHC-PMS

A medical ethics manager who must ensure that the system meets current ethical guidelines for patient care. Health care managers who obtain management information from the system.

- A medical ethics manager who must ensure that the system meets current ethical guidelines for patient care.
- Health care managers who obtain management information from the system.
- Medical records staff who are responsible for ensuring that system information can be maintained and preserved, and that record keeping procedures have been properly implemented.

Keywords: Stakeholders, Health, Medical

Interviewing

Formal or informal interviews with stakeholders are part of most RE processes.

Types of interview Closed interviews based on pre-determined list of questions Open interviews where various issues are explored with stakeholders. Prompt the interviewee to get discussions going using a springboard question, a requirements proposal, or by working together on a prototype system.

- Types of interview
- Closed interviews based on pre-determined list of questions
- Open interviews where various issues are explored with stakeholders.
- Effective interviewing
- Be open-minded, avoid pre-conceived ideas about the requirements and are willing to listen to stakeholders.
- Prompt the interviewee to get discussions going using a springboard question, a requirements proposal, or by working together on a prototype system.

Keywords: Interviewing Formal, Types, Closed, Open, Effective, Prompt

Interviews in practice

Normally a mix of closed and open-ended interviewing.

Interviews are good for getting an overall understanding of what stakeholders do and how they might interact with the system. Interviews are not good for understanding domain requirements Requirements engineers cannot understand specific domain terminology; Some domain knowledge is so familiar that people find it hard to articulate or think that it isn't worth articulating.

- Interviews are good for getting an overall understanding of what stakeholders do and how they might interact with the system.
- Interviews are not good for understanding domain requirements
- Requirements engineers cannot understand specific domain terminology;
- Some domain knowledge is so familiar that people find it hard to articulate or think that it isn't worth articulating.

Keywords: Interviews, Normally, Requirements, Some

Scenarios

Scenarios are real-life examples of how a system can be used.

They should include A description of the starting situation; A description of the normal flow of events; A description of what can go wrong; Information about other concurrent activities; A description of the state when the scenario finishes.

- They should include
- A description of the starting situation;
- A description of the normal flow of events;
- A description of what can go wrong;
- Information about other concurrent activities;
- A description of the state when the scenario finishes.

Keywords: Scenarios Scenarios, They, Information

Use cases

High-level graphical model supplemented by more detailed tabular description (see Chapter 5). Sequence diagrams may be used to add detail to use-cases by showing the sequence of event processing in the system.

- Use-cases are a scenario based technique in the UML which identify the actors in an interaction and which describe the interaction itself.
- A set of use cases should describe all possible interactions with the system.
- High-level graphical model supplemented by more detailed tabular description (see Chapter 5).
- Sequence diagrams may be used to add detail to use-cases by showing the sequence of event processing in the system.

Keywords: High, Chapter, Sequence

Ethnography

A social scientist spends a considerable time observing and analysing how people actually work. Ethnographic studies have shown that work is usually richer and more complex than suggested by simple system models.

- A social scientist spends a considerable time observing and analysing how people actually work.
- People do not have to explain or articulate their work.
- Social and organisational factors of importance may be observed.
- Ethnographic studies have shown that work is usually richer and more complex than suggested by simple system models.

Keywords: Ethnography, People, Social, Ethnographic

Scope of ethnography

Requirements that are derived from the way that people actually work rather than the way I which process definitions suggest that they ought to work. Requirements that are derived from cooperation and awareness of other people's activities.

- Requirements that are derived from the way that people actually work rather than the way I which process definitions suggest that they ought to work.
- Requirements that are derived from cooperation and awareness of other people's activities.
- Awareness of what other people are doing leads to changes in the ways in which we do things.
- Ethnography is effective for understanding existing processes but cannot identify new features that should be added to a system.

Keywords: Scope, Requirements, Awareness, Ethnography

Focused ethnography

Developed in a project studying the air traffic control process

Combines ethnography with prototyping Prototype development results in unanswered questions which focus the ethnographic analysis. The problem with ethnography is that it studies existing practices which may have some historical basis which is no longer relevant.

- Combines ethnography with prototyping
- Prototype development results in unanswered questions which focus the ethnographic analysis.
- The problem with ethnography is that it studies existing practices which may have some historical basis which is no longer relevant.

Keywords: Focused, Developed, Combines, Prototype

Requirements validation

Concerned with demonstrating that the requirements define the system that the customer really wants. Requirements error costs are high so validation is very important Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error.

- Concerned with demonstrating that the requirements define the system that the customer really wants.
- Requirements error costs are high so validation is very important
- Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error.

Keywords: Requirements, Concerned, Fixing

Requirements checking

Validity. Does the system provide the functions which best support the customer’s needs?

Consistency. Are there any requirements conflicts?

- Consistency. Are there any requirements conflicts?
- Completeness. Are all functions required by the customer included?
- Realism. Can the requirements be implemented given available budget and technology
- Verifiability. Can the requirements be checked?

Keywords: Requirements, Validity, Does, Consistency, Completeness, Realism, Verifiability

Requirements validation techniques

Requirements reviews

Systematic manual analysis of the requirements. Test-case generation Developing tests for requirements to check testability.

- Systematic manual analysis of the requirements.
- Prototyping
- Using an executable model of the system to check requirements. Covered in Chapter 2.
- Test-case generation
- Developing tests for requirements to check testability.

Keywords: Requirements, Systematic, Prototyping Using, Covered, Chapter, Test, Developing

Requirements reviews

Regular reviews should be held while the requirements definition is being formulated.

Both client and contractor staff should be involved in reviews. Good communications between developers, customers and users can resolve problems at an early stage.

- Both client and contractor staff should be involved in reviews.
- Reviews may be formal (with completed documents) or informal. Good communications between developers, customers and users can resolve problems at an early stage.

Keywords: Requirements, Regular, Both, Reviews, Good

Review checks

Verifiability

Is the requirement realistically testable? Comprehensibility Is the requirement properly understood?

- Is the requirement realistically testable?
- Comprehensibility
- Is the requirement properly understood?
- Traceability
- Is the origin of the requirement clearly stated?
- Adaptability
- Can the requirement be changed without a large impact on other requirements?

Keywords: Review, Verifiability Is, Comprehensibility Is, Traceability Is, Adaptability Can

Requirements management

You need to keep track of individual requirements and maintain links between dependent requirements so that you can assess the impact of requirements changes. You need to establish a formal process for making change proposals and linking these to system requirements.

- Requirements management is the process of managing changing requirements during the requirements engineering process and system development.
- New requirements emerge as a system is being developed and after it has gone into use.

Keywords: Requirements

Changing requirements

The business and technical environment of the system always changes after installation.

New hardware may be introduced, it may be necessary to interface the system with other systems, business priorities may change (with consequent changes in the system support required), and new legislation and regulations may be introduced that the system must necessarily abide by. System customers impose requirements because of organizational and budgetary constraints.

- The people who pay for a system and the users of that system are rarely the same people.

Keywords: Changing, System

Changing requirements

Large systems usually have a diverse user community, with many users having different requirements and priorities that may be conflicting or contradictory. The final system requirements are inevitably a compromise between them and, with experience, it is often discovered that the balance of support given to different users has to be changed.

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- The final system requirements are inevitably a compromise between them and, with experience, it is often discovered that the balance of support given to different users has to be changed.

Keywords: Changing, Large

Requirements management planning

Establishes the level of requirements management detail that is required.

Requirements management decisions: Requirements identification Each requirement must be uniquely identified so that it can be cross-referenced with other requirements. Tool support Tools that may be used range from specialist requirements management systems to spreadsheets and simple database systems.

- Requirements management decisions:
- Requirements identification Each requirement must be uniquely identified so that it can be cross-referenced with other requirements.
- A change management process This is the set of activities that assess the impact and cost of changes. I discuss this process in more detail in the following section.
- Traceability policies These policies define the relationships between each requirement and between the requirements and the system design that should be recorded.
- Tool support Tools that may be used range from specialist requirements management systems to spreadsheets and simple database systems.

Keywords: Requirements, Establishes, Each, Traceability, Tool, Tools

Requirements change management

Deciding if a requirements change should be accepted

This analysis is fed back to the change requestor who may respond with a more specific requirements change proposal, or decide to withdraw the request. The effect of the proposed change is assessed using traceability information and general knowledge of the system requirements.

- Problem analysis and change specification
- Change analysis and costing
- Change implementation
- The requirements document and, where necessary, the system design and implementation, are modified. Ideally, the document should be organized so that changes can be easily implemented.

Keywords: Requirements, Deciding, During, Once, Problem, Change, Ideally

Key points

Requirements validation is the process of checking the requirements for validity, consistency, completeness, realism and verifiability. Business, organizational and technical changes inevitably lead to changes to the requirements for a software system.

- You can use a range of techniques for requirements elicitation including interviews, scenarios, use-cases and ethnography.
- Requirements validation is the process of checking the requirements for validity, consistency, completeness, realism and verifiability.
- Business, organizational and technical changes inevitably lead to changes to the requirements for a software system. Requirements management is the process of managing and controlling these changes.

Keywords: Requirements, Business